Soft Prior or Hard Mask

How the nine gene-gene graphs enter the cell graph transformer: the graph-regularization sweep on the 025 triples, read at three seeds per arm

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Abstract

Nine gene-gene graphs enter layer 1 of the cell graph transformer by one of two mechanisms: a soft prior, a divergence penalty of weight λ that pulls each head's attention toward its graph's normalized adjacency, or a hard mask that restricts each head's attention to its graph's edges. The sweep runs one protocol (the 376,732 triples of experiment 010 on their pinned random split, a learnable gene table, AdamW at a constant 2.5×10^{-4}, 30 epochs, four A40s per run) at $\lambda \in \{0, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100\}$, under the mask, and under the prior toward degree-matched random rewirings of the same graphs, three seeds per arm. Read at epoch 29, the mask sits with no penalty, 0.421 ± 0.003 against 0.425 ± 0.000 held-out Pearson, and the prior rises with λ to 0.447 ± 0.001 at $\lambda = 1$, above no penalty on every seed by 0.021 to 0.023, and holds there for two more decades (0.444 ± 0.008 at 10, 0.443 ± 0.004 at 100, above no penalty on every seed) with no turnover in reach. Read at each run's best epoch, all eleven arms lie within 0.010 of one another (0.444 to 0.454): the prior moves the peak by at most 0.006 and moves when it occurs, from epoch 15 or 16 without a penalty to epochs 23 to 26 at $\lambda = 1$, so most of what it buys at the fixed reading is resistance to the decline the unpenalized model shows after epoch 16. The attention contracts onto the graph at every λ tested, with edge recall at degree 0.78 at 10^{-5} and 0.915 from 10^{-2} up against 0.006 with no penalty, and saturates two decades below the λ at which accuracy is highest, so the contraction itself is not what carries the accuracy. The random-graph control, a prior of the same weight toward degree-matched rewirings of the nine graphs, sits at the no-penalty level at epoch 29 at every λ tested (0.420 ± 0.012 at 10^{-3}, three seeds; 0.430 ± 0.006 at 10^{-1} and 0.427 ± 0.004 at 1, five seeds each, against 0.425), where the biological graphs give 0.432, 0.445 and 0.447; paired by seed the biological graphs are above the random ones on every seed at 10^{-1} and 1, by 0.017 at epoch 29 ($p = 0.005$ and 0.013) and 0.015 to 0.018 at the best epoch ($p = 0.03$ and 0.014), and at the best epoch the random arms fall below no penalty (-0.011 to -0.012, $p = 0.026$ and 0.007). The hypothesis under test, that the graphs help only when the model is optimized through the divergence and not when they are imposed as a constraint, holds on the fixed-epoch reading, and the alternative that any graph of the same degree sequence would do the same is rejected at the two λ where the gain is largest. The follow-up rounds (Section 6) measure the rest: a two- or three-hop target or mask does not beat one hop; a symmetric target or a directed mask moves neither mechanism toward the other; the prior on layers 3 to 4 equals the prior on layer 1 while every mask placement stays at no penalty; at 60 epochs the unpenalized model keeps declining (0.391 to 0.405 at epoch 59) and the prior at $\lambda = 1$ holds (0.433 to 0.437); with the composite sequence embedding the model does not train at all without the prior and trains slowly with it; and the width-360 arms did not run, out of GPU memory on the A40.

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1 The question ✗

The cell graph transformer of experiment 010 attends over all 6,607 genes in every layer. Nine gene-gene graphs, physical and regulatory interactions, TFLink, and the six STRING channels (neighborhood, fusion, co-occurrence, co-expression, experimental, database), are given to nine heads of layer 1, one graph per head. Two mechanisms can hand a graph to a head. The **soft prior** adds to the training loss, for each head k with row-normalized adjacency $\tilde{A}_k$ and attention α_k on the wild-type graph,

$$\lambda \sum_{k=1}^{9} \frac{N}{|E_k|} \sum_i D_{\mathrm{KL}}\big(\tilde{A}_{k,i\cdot} \,\|\, \alpha_{k,i\cdot}\big)\,, \tag{1}$$

with N the number of genes and $|E_k|$ the edge count of graph k, so a head is pulled toward spreading each gene's attention uniformly over that gene's neighbors and is free to put mass anywhere. The rest of the loss is the same in every arm: the mean squared error of the z-scored interaction (the point loss, weight 1) and a Sinkhorn-regularized Wasserstein distance between the batch's predicted and observed label distributions (weight 0.1, blur 0.05, extttgeomloss); the gradient probe shows the Wasserstein term at about a fifth of the point-loss gradient after epoch 0 and it is not varied here. The **hard mask** sets the attention logits of every non-edge (self-loops kept) to $-\infty$ before the softmax in layer 1, so a head can only attend along the graph and needs no penalty and no λ. The two are the ends of one axis: the mask is the $\lambda \to \infty$ limit of Eq. 1 on the support, but not in the weights, since the mask lets a head weight its neighbors as it likes while the prior pulls toward the uniform row of $\tilde{A}_k$.

Earlier readings put the two mechanisms far apart. The 010 model at $\lambda = 10^{-3}$ under a cosine schedule reached 0.456 to 0.464 held-out Pearson over three seeds, and on the 025 build the same configuration replicated at 0.446 (GilaHyper job 1598, W&B `0yw7moue`), while the two hard-mask arms run then collapsed: 0.307 at epoch 1 and then near 0 (`7f1yrsq9`), or near 0 for 17 epochs (`4qmgkcgn`). Those mask runs were under the cosine schedule with a different normalizer, so the collapse was not a controlled comparison. The hypothesis this sweep was designed to test is stated in the form it was proposed: *the graphs help, but only when the model is optimized through the divergence; imposing them as a mask does not help.* A second question rides on the first: if the prior helps, is it the biology of the nine graphs or would any graph with the same degree sequence condition training the same way?

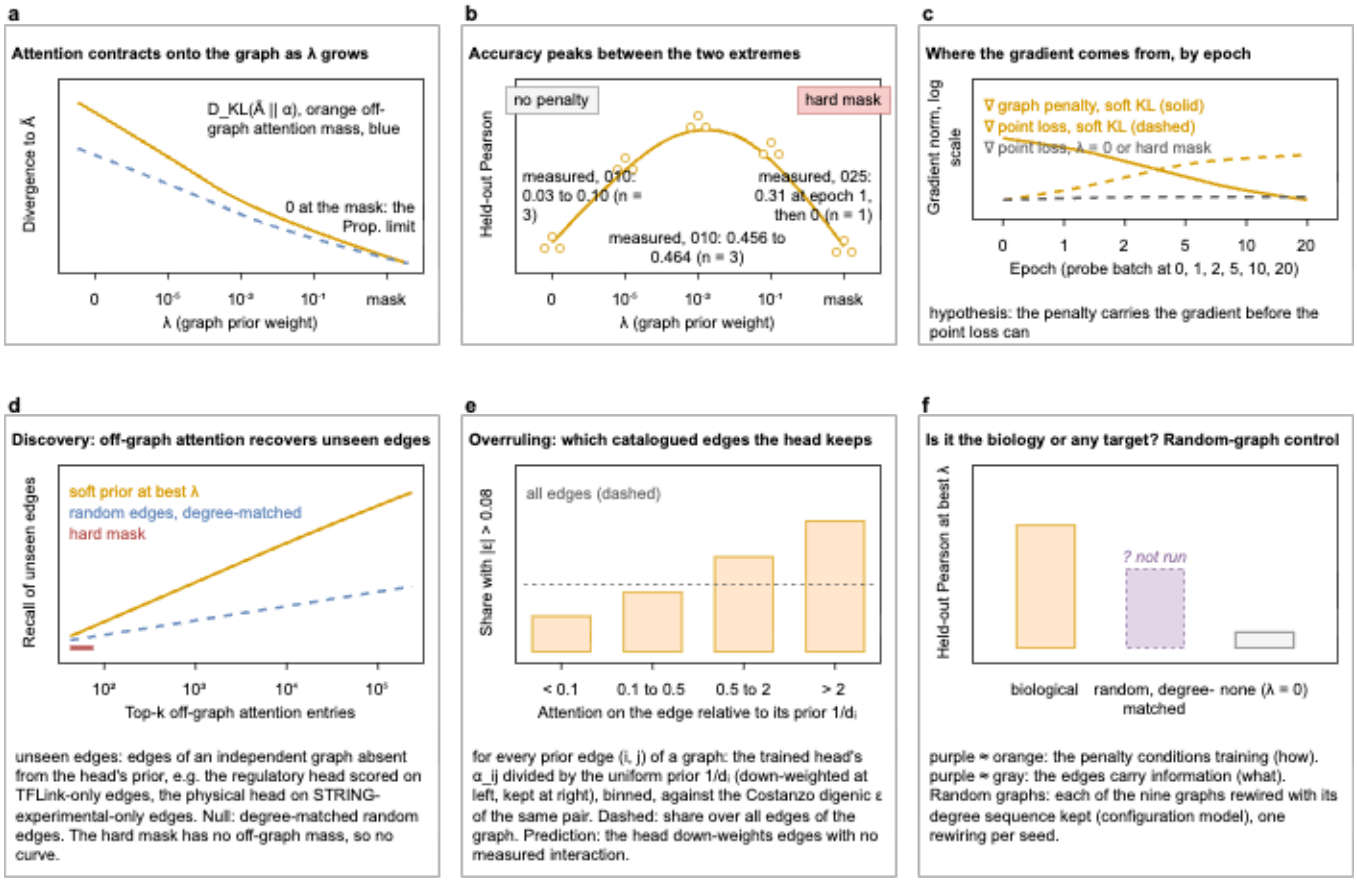
Design. Every arm is the constant-rate control `cgt_s0_r_kl_ctrl_013` with one key changed (Table 1). The control is the 376,732 triples of experiment 010 (subset S0 of the 025 build) on 010's pinned random split, 301,386 training and 37,673 validation triples, a learnable 1,000-dimensional gene table, the label normalizer fit on the training split, AdamW at 2.5×10^{-4} with no schedule, batch 256 over four GPUs, 30 epochs. The runs are Delta `gpuA40x4` jobs, one node and four A40s per run, 24 h clocks, the 554 GB LMDB staged onto node-local NVMe at the start of each job (`experiments/025-solid-growth/scripts/delta_cgt.slurm`, `experiments/025-solid-growth/scripts/delta_submit_sweep.sh`; jobs 22055147 to 22055169, 22056267 to 22056269, 22080055, 22080057 and 22107791). The prior arms peak at 6.05 GB of allocated GPU memory per card, the mask at 2.76 GB, because only the penalized layer materializes its attention.

Table 1: The arms. Every arm is `cgt_s0_r_kl_ctrl_013` with the one change named; the control itself is the ladder's $\lambda = 10^{-3}$ point. Complete runs are those with 30 logged epochs on W&B; at the pull of 2026-10-10 every seed listed is complete (the $\lambda = 10$ and 100 arms, the second seed of 10^{-5} and the random arms at 10^{-1} and 1 were added in round 1b, Section 6).

arm	change on the control	graph target	complete seeds
no penalty	`graph_reg_lambda=0`	none	1, 2, 3
prior ladder	`graph_reg_lambda` $\in \{10^{-5}, 10^{-4}, 10^{-2}, 10^{-1}, 1, 10, 100\}$	nine biological graphs, layer 1	1, 2, 3 each
control	none	nine biological graphs, layer 1, $\lambda = 10^{-3}$	1, 2, 3
hard mask	`cgt_s0_r_mask_028`: mask on, $\lambda = 0$	nine biological graphs, layer 1	1, 2, 3
random graphs	`cgt_s0_r_kl_rand_031`: rewiring on	nine rewired graphs, layer 1, $\lambda = 10^{-3}$; 10^{-1} and 1	1, 2, 3; 1 to 5 each

Readings. Every run is read three times. The **fixed-epoch reading** is validation Pearson at epoch 29, the protocol's last epoch, the same optimization state for every arm. The **max-over-epochs reading** is the largest validation Pearson over the 30 epochs, an upward-biased order statistic reported with the epoch at which it occurred; it is the number a best-checkpoint deployment would see. The **minimum-loss reading** is validation Pearson at the epoch of that run's lowest validation point loss, the epoch a loss-monitored checkpoint would keep; the loss curve is jagged after its minimum, so the epoch it picks varies by seed. Arm values are the mean and sample standard deviation over seeds; a paired comparison is by seed, since the split is pinned and the seed sets the initialization and the batch order for every arm alike.

Vocabulary. Terms used in the figures and tables, in alphabetical order.



Experiment design (Delta, gpuA40x4, 4-GPU DDP, one node per run) MOCK-UP: sketched curves, not data

Arm	λ	Graph target	Seeds	Runs	Wall clock	Panels
no penalty	0	none	3	3	12 h	b, c, f
λ sweep	10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}	9 biological graphs, layer 1	3 each	15	24 h	a, b, c, d, e
hard mask	∞ (mask at layer 1)	9 biological graphs	3	3	12 h	b, c, d
random-graph control	best λ from the sweep	degree-matched random	3	3	24 h	f
total				24		2,016 GPU-h

Fixed across arms: experiment-025 S0 (the 376,732 records of 010), the pinned 010 random split, 010's schedule (batch 256, bf16, OneCycle to 5e-4), one regularized layer (layer 1, all nine heads), each run read at its best-validation checkpoint. GPU-hours charge per GPU: 4 x wall clock per run. Anchors already measured: 010 λ = 0 and λ = 10^{-3} (x367 coefficient), 025 soft KL and hard mask. Panels d and e can be computed today on the three 010 checkpoints, before any run.

Figure 1: The figure as it was designed before any run, drawn in draw.io from `notes/assets/drawio/FigS-graph-regularization-sweep.gen.py` before the sweep was submitted (week of 2026-09-08): sketched curves, not data. Panel a predicted contraction of attention onto the graph with λ; b an accuracy peak between no penalty and the mask; c the penalty carrying the gradient before the point loss can; d discovery of unseen edges from off-graph attention; e overruling of cataloged edges by the digenic interaction; f the degree-matched random-graph control at the best λ. The design strip below the panels lists the arms and the Delta budget under the cosine schedule that the constant-rate protocol later replaced. Figure 2 is what was measured; Section 5 sets the two side by side.

Table 2: Controlled vocabulary.

term	meaning
divergence	Eq. 1 without its λ: the logged penalty divided by λ, summed over the nine heads and normalized by edge count, computed on the wild-type graph. Comparable across the ladder; not logged at $\lambda = 0$ or under the mask, where the model skips the computation. Reported from the validation pass at epoch 29.
edge recall at degree	For each gene i of graph k with degree d_i, the fraction of i's neighbors among the d_i largest entries of $\alpha_{k,i,\cdot}$; averaged over genes, then over the nine heads. It is 1 by construction under the mask and near $1/N$ for attention unrelated to the graph. Logged at epochs 9, 19 and 29.
gradient probe	On the first batch of epochs 0, 1, 2, 5, 10 and 20, one extra backward pass per loss term; the ℓ_2 norm of the parameter gradient of the point loss and of the penalty, before clipping, and their ratio.
point loss	Mean squared error of the prediction against the z-scored interaction label.
random graphs, degree-matched	Each of the nine graphs rewired by seeded double-edge swaps (five swaps per edge) that keep every gene's degree, in and out for the directed graphs; the rewiring seed is the run seed. The prior and the edge diagnostics are computed against the rewired graphs.
S0	The 376,732 measured triples of experiment 010, as stored in the 025 build; the pool of every arm here.

2 The ladder: accuracy against the prior's weight ✗

Table 3: Held-out Pearson by arm, mean over seeds (± sample sd) at the pull of 2026-10-10, every arm complete, at the three readings: epoch 29, the max over epochs, and the epoch of minimum validation point loss. The highest value in each reading is bold; an arrow marks an arm whose seeds are all above (↑) or all below (↓) no penalty on the same reading. Generated by `graph_reg_sweep_readout.py`.

arm	n	Pearson, epoch 29	Pearson, max over epochs	Pearson, at min val loss
no penalty ($\lambda = 0$)	3	0.425 ± 0.000	0.448 ± 0.002	0.440 ± 0.005
KL $\lambda = 10^{-5}$	3	0.422 ± 0.005	0.445 ± 0.003	0.442 ± 0.003
KL $\lambda = 10^{-4}$	3	0.419 ± 0.010	0.446 ± 0.003	0.442 ± 0.001
KL $\lambda = 10^{-3}$ (control)	3	0.432 ± 0.005 ↑	0.451 ± 0.004 ↑	0.444 ± 0.003
KL $\lambda = 10^{-2}$	3	0.438 ± 0.002 ↑	0.453 ± 0.002 ↑	0.447 ± 0.005
KL $\lambda = 10^{-1}$	3	0.445 ± 0.009 ↑	**0.454 ± 0.006**	**0.453 ± 0.006** ↑
KL $\lambda = 1$	3	**0.447 ± 0.001** ↑	0.452 ± 0.003 ↑	0.443 ± 0.009
hard mask, layer 1	3	0.421 ± 0.003 ↓	0.446 ± 0.003	0.442 ± 0.003
KL $\lambda = 10^{-3}$, random graphs	3	0.420 ± 0.012	0.447 ± 0.002	0.444 ± 0.006
KL $\lambda = 10$	3	0.444 ± 0.008 ↑	0.452 ± 0.004 ↑	0.442 ± 0.005
KL $\lambda = 100$	3	0.443 ± 0.004 ↑	0.451 ± 0.009	0.440 ± 0.004
KL $\lambda = 10^{-1}$, random graphs	5	0.430 ± 0.006	0.437 ± 0.003 ↓	0.434 ± 0.004
KL $\lambda = 1$, random graphs	5	0.427 ± 0.004 ↑	0.435 ± 0.004 ↓	0.430 ± 0.006 ↓
hard mask, directed	3	0.416 ± 0.012 ↓	0.448 ± 0.001	0.443 ± 0.002
hard mask, 2 hops, directed	3	0.419 ± 0.006	0.442 ± 0.003 ↓	0.440 ± 0.001
hard mask, 3 hops, directed	3	0.417 ± 0.003 ↓	0.445 ± 0.003 ↓	0.439 ± 0.004 ↓
KL $\lambda = 1$, 2-hop target	3	0.438 ± 0.006 ↑	0.442 ± 0.004 ↓	0.437 ± 0.004 ↓
KL $\lambda = 1$, 3-hop target	3	0.429 ± 0.012	0.442 ± 0.006	0.434 ± 0.004 ↓
KL $\lambda = 1$, symmetric target	3	0.439 ± 0.010 ↑	0.449 ± 0.003	0.446 ± 0.007
hard mask, layers 1–2	3	0.419 ± 0.004 ↓	0.446 ± 0.001 ↓	0.440 ± 0.004
hard mask, layers 3–4	3	0.415 ± 0.007 ↓	0.445 ± 0.001 ↓	0.442 ± 0.002
hard mask, layers 1–4	3	0.426 ± 0.009	0.446 ± 0.002	0.445 ± 0.003
KL $\lambda = 1$, layers 1–2	3	0.437 ± 0.009 ↑	0.446 ± 0.006	0.440 ± 0.007
KL $\lambda = 1$, layers 3–4	3	0.445 ± 0.007 ↑	0.449 ± 0.004	0.441 ± 0.011
no penalty, 60 epochs	3	0.415 ± 0.003	0.443 ± 0.004	0.440 ± 0.001
hard mask, 60 epochs	3	0.419 ± 0.009	0.446 ± 0.001 ↑	0.440 ± 0.004
KL $\lambda = 1$, 60 epochs	3	0.441 ± 0.005 ↑	0.450 ± 0.006	0.445 ± 0.011
no penalty, composite embedding	3	-0.006 ± 0.004	0.073 ± 0.061	-0.002 ± 0.001
KL $\lambda = 1$, composite embedding	3	0.421 ± 0.007 ↑	0.423 ± 0.007 ↑	0.423 ± 0.007 ↑

The mask is no penalty. On the fixed-epoch reading the hard mask gives 0.421 ± 0.003 and no penalty 0.425 ± 0.000 (Table 3); by seed the mask is below no penalty on all three, by 0.001, 0.007 and 0.003. On the max-over-epochs reading the two are 0.446 ± 0.003 and 0.448 ± 0.002, the mask below on two seeds of three. The mask arm also fits the training triples least, 0.612 against 0.624 train Pearson at epoch 29 (Table 4). Restricting the support of layer-1 attention to the nine graphs therefore changes nothing a validation triple can see. The collapse of the two earlier mask runs did not recur under this protocol: all three seeds train normally (Figure 2g).

The prior rises with λ and does not peak inside the ladder. At epoch 29 the ladder reads 0.422, 0.419, 0.432, 0.438, 0.445, 0.447, 0.444 and 0.443 at $\lambda = 10^{-5}$ through 100 (Table 3). The two smallest weights sit at no penalty, mixed by seed (-0.003, $+0.002$, -0.007 at 10^{-5}); from 10^{-3} up every seed is above its no-penalty partner, by 0.005 to 0.013 at 10^{-3} and by 0.021, 0.023 and 0.021 at $\lambda = 1$. The mock-up drawn before the runs (`notes/assets/drawio/FigS-graph-regularization-sweep.gen.py`) placed a peak between the two extremes; the measurement places the maximum at $\lambda = 1$, with 10^{-1} and 1 within one standard deviation of each other (0.445 ± 0.009, 0.447 ± 0.001), and the two decades above it, run in round 1b, do not turn over: 0.444 ± 0.008 at $\lambda = 10$ and 0.443 ± 0.004 at 100, above no penalty on every seed ($+0.013$ to $+0.029$, $p = 0.056$; $+0.015$ to $+0.023$, $p = 0.019$) and within one standard deviation of $\lambda = 1$. Divergence (295 to 304) and edge recall (0.915 to 0.916) do not move from $\lambda = 1$ to 100 either, while the penalty's share of the gradient rises from 10^3 to 10^5 times the point loss (Figure 2d, e): from 10^{-1} up the prior is saturated and the accuracy is flat, so there is no weight at which the soft prior turns into the mask's result.

The max-over-epochs reading compresses the ladder. Read at each run's best epoch, the eleven arms span 0.444 to 0.454, and the best arm ($\lambda = 10^{-1}$, 0.454 ± 0.006) is 0.006 above no penalty (0.448 ± 0.002). At $\lambda = 1$ the paired differences against no penalty shrink from 0.021 to 0.023 at epoch 29 to 0.007, 0.003 and 0.003 at the best epoch, still positive on all three seeds. What changes with λ is less the height of the peak than its timing: the no-penalty seeds peak at epochs 15, 16 and 16, the mask at 15, 17 and 18, $\lambda = 1$ at 23, 24 and 26 (Table 6). The fixed-epoch gain of the prior is therefore mostly the unpenalized model's decline from epoch 16 to epoch 29 (0.441 to 0.425 in the seed mean) that the prior at high λ does not show (0.435 at 15, 0.447 at 29).

The loss-selected checkpoint flattens the ladder. A loss-monitored checkpoint keeps the epoch of minimum

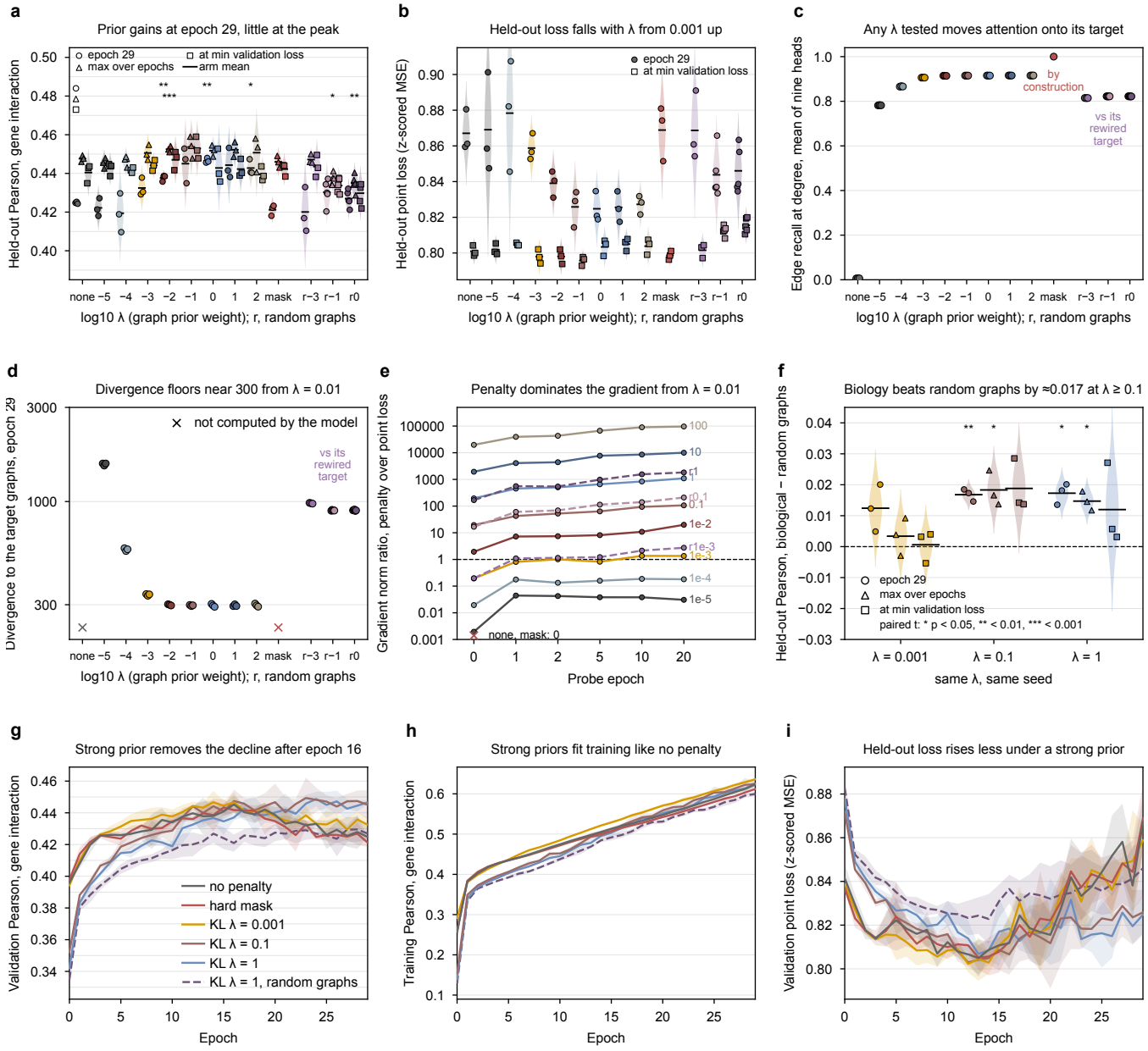


Figure 2: The sweep at three seeds per arm, read from W&B by `experiments/025-solid-growth/scripts/graph_reg_sweep_readout.py` and drawn by `experiments/025-solid-growth/scripts/graph_reg_sweep_plots.py`. Each arm keeps one color in every panel; the ladder axis names the arms, and the random-graph arms are dashed in **e** and **g**–**i**. Every point is one seed, filled in its arm's color with a black edge; a short black bar is the arm mean; in **a**–**c** and **f** the light shape behind a column is a normal density with that column's seed mean and standard deviation, a reading aid at three seeds rather than an estimate of the distribution; a cross marks a quantity the model does not compute. The ladder axis is $\log_{10}\lambda$ from -5 to 2, with *none* the unpenalized arm, *mask* the hard mask and *r−3*, *r−1* and *r0* the degree-matched random-graph arms at $\lambda = 10^{-3}$, 10^{-1} and 1 (three, five and five seeds). **a**, Held-out Pearson on the 37,673 validation triples at three readings per seed: epoch 29 (circle, left), max over epochs (triangle, middle, every run) and at the epoch of minimum validation point loss (square, right). Stars are the paired t of each reading against no penalty on the same seeds, in a band above the data with one row per reading, the row's marker at its left end (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$); the minimum-loss reading has none, its closest being $\lambda = 10^{-1}$ at $p = 0.08$. **b**, Held-out point loss at epoch 29 (circle) and its minimum over epochs (square), the value at the epoch the square in **a** is read at. **c**, Edge recall at degree at epoch 29, mean over the nine heads, each arm against its own target: the random arm is scored against the rewired graphs it was trained toward, the mask is 1 by construction. **d**, Divergence of layer-1 attention from the target graphs at epoch 29, validation pass, log scale; not computed at $\lambda = 0$ or under the mask. **e**, Gradient norm of the penalty over that of the point loss on the probe batch at epochs 0, 1, 2, 5, 10 and 20, mean over seeds, log scale; the two arms without a penalty are at 0. **f**, The control as a paired difference: biological minus degree-matched random graphs, seed by seed on the three seeds both arms share, at each λ the random arm runs at, under the three readings of **a** (same markers, arm mean as a bar); stars are the paired t of that difference. **a** gives the levels and **f** the pairing across arms, which **a** does not show. **g**, Validation Pearson, **h**, training Pearson and **i**, validation point loss by epoch, mean over complete seeds with a ± sd band. Panel titles state the finding each panel shows and are checked against the tables in Section 5.

Table 4: Diagnostics by arm, mean over complete seeds (± sample sd): validation point loss at epoch 29 and at its minimum with the mean epoch of that minimum, training Pearson at epoch 29, the divergence of layer-1 attention from the arm's target graphs (validation pass, epoch 29), and edge recall at degree against the same target, mean over the nine heads. Generated by `graph_reg_sweep_readout.py`.

arm	point loss, ep 29	min point loss (epoch)	train Pearson, ep 29	divergence	edge recall
no penalty ($\lambda = 0$)	0.867 ± 0.012	0.801 ± 0.003 (12.0)	0.624 ± 0.006	–	0.006 ± 0.000
KL $\lambda = 10^{-5}$	0.869 ± 0.028	0.802 ± 0.003 (12.0)	0.658 ± 0.001	1546	0.782 ± 0.001
KL $\lambda = 10^{-4}$	0.878 ± 0.031	0.805 ± 0.001 (7.7)	0.663 ± 0.007	571	0.866 ± 0.001
KL $\lambda = 10^{-3}$ (control)	0.859 ± 0.007	0.798 ± 0.004 (12.0)	0.636 ± 0.004	337	0.906 ± 0.000
KL $\lambda = 10^{-2}$	0.839 ± 0.007	0.798 ± 0.004 (15.3)	0.613 ± 0.006	300	0.914 ± 0.000
KL $\lambda = 10^{-1}$	0.826 ± 0.010	0.796 ± 0.003 (17.3)	0.625 ± 0.006	297	0.915 ± 0.000
KL $\lambda = 1$	0.825 ± 0.009	0.803 ± 0.004 (18.7)	0.624 ± 0.008	298	0.915 ± 0.000
hard mask, layer 1	0.869 ± 0.015	0.799 ± 0.002 (14.0)	0.612 ± 0.001	–	–
KL $\lambda = 10^{-3}$, random graphs	0.869 ± 0.020	0.802 ± 0.004 (12.3)	0.621 ± 0.004	980	0.814 ± 0.000
KL $\lambda = 10$	0.826 ± 0.009	0.805 ± 0.003 (14.7)	0.616 ± 0.007	297	0.915 ± 0.000
KL $\lambda = 100$	0.827 ± 0.005	0.804 ± 0.004 (15.7)	0.586 ± 0.002	298	0.915 ± 0.000
KL $\lambda = 10^{-1}$, random graphs	0.844 ± 0.013	0.813 ± 0.003 (15.4)	0.610 ± 0.007	902	0.822 ± 0.000
KL $\lambda = 1$, random graphs	0.846 ± 0.013	0.816 ± 0.004 (15.0)	0.600 ± 0.006	904	0.822 ± 0.000
hard mask, directed	0.869 ± 0.014	0.799 ± 0.003 (14.0)	0.607 ± 0.010	–	–
hard mask, 2 hops, directed	0.863 ± 0.017	0.802 ± 0.003 (13.3)	0.603 ± 0.004	–	–
hard mask, 3 hops, directed	0.868 ± 0.009	0.803 ± 0.003 (14.0)	0.607 ± 0.014	–	–
KL $\lambda = 1$, 2-hop target	0.831 ± 0.008	0.806 ± 0.001 (16.0)	0.621 ± 0.007	60	0.924 ± 0.000
KL $\lambda = 1$, 3-hop target	0.848 ± 0.012	0.809 ± 0.003 (11.7)	0.632 ± 0.006	31	0.959 ± 0.000
KL $\lambda = 1$, symmetric target	0.828 ± 0.018	0.805 ± 0.002 (24.0)	0.602 ± 0.007	433	0.829 ± 0.000
hard mask, layers 1–2	0.864 ± 0.022	0.802 ± 0.002 (11.3)	0.610 ± 0.009	–	–
hard mask, layers 3–4	0.870 ± 0.014	0.800 ± 0.001 (13.3)	0.620 ± 0.009	–	–
hard mask, layers 1–4	0.849 ± 0.018	0.796 ± 0.002 (12.0)	0.613 ± 0.004	–	–
KL $\lambda = 1$, layers 1–2	0.839 ± 0.015	0.806 ± 0.005 (17.0)	0.617 ± 0.001	649	0.914 ± 0.000
KL $\lambda = 1$, layers 3–4	0.822 ± 0.019	0.807 ± 0.005 (20.7)	0.629 ± 0.002	725	–
no penalty, 60 epochs	0.880 ± 0.017	0.801 ± 0.001 (14.0)	0.606 ± 0.002	–	0.006 ± 0.000
hard mask, 60 epochs	0.861 ± 0.017	0.801 ± 0.002 (10.3)	0.606 ± 0.008	–	–
KL $\lambda = 1$, 60 epochs	0.831 ± 0.010	0.801 ± 0.006 (17.0)	0.621 ± 0.008	298	0.915 ± 0.000
no penalty, composite embedding	0.993 ± 0.001	0.992 ± 0.000 (20.3)	0.001 ± 0.002	–	0.006 ± 0.001
KL $\lambda = 1$, composite embedding	0.836 ± 0.006	0.821 ± 0.005 (26.7)	0.437 ± 0.008	2510	0.762 ± 0.001

validation point loss, which falls at epoch 12 with no penalty and drifts later with λ (15 at 10^{-2}, 17 at 10^{-1}, 19 at $\lambda = 1$; Table 4). Read there, no penalty gives 0.440 ± 0.005, the mask 0.442 ± 0.003, and the ladder 0.442, 0.442, 0.444, 0.447, 0.453 ± 0.006 and 0.443 ± 0.009 from 10^{-5} to 1 (Table 3). Only $\lambda = 10^{-1}$ stands above no penalty on every seed ($+0.012$, $p = 0.08$); $\lambda = 1$ does not, because its loss minimum lands late on a jagged curve (Figure 2i) at epochs where its Pearson is 0.435 to 0.452. The reading a deployment would see therefore depends on the stopping rule: a fixed budget favors the strongest prior, a loss-selected checkpoint favors 10^{-1}, and a Pearson-selected checkpoint separates nothing.

Paired by seed. Table 5 gives each arm minus no penalty on the same seed. With three seeds the paired t has two degrees of freedom and is reported for what it is: at epoch 29 the prior at $\lambda = 1$ is above no penalty by $+0.022 \pm 0.001$ ($t = 30$, $p = 0.001$) and at 10^{-2} by $+0.013 \pm 0.001$ ($t = 18$, $p = 0.003$); 10^{-1} is above on every seed but with a wider spread ($+0.020 \pm 0.010$, $p = 0.07$). The mask is below no penalty on every seed (-0.004 ± 0.003, $p = 0.16$). At the best-epoch reading only 10^{-2} separates ($+0.005 \pm 0.000$, $p < 0.001$); $\lambda = 1$ is above on every seed by $+0.004 \pm 0.002$ ($p = 0.09$) and the mask is mixed. The claim the fixed-epoch reading supports at three seeds is that a strong prior beats no penalty by about 0.02; the claim it does not support is that any arm's peak is higher than another's by more than 0.005. The model class has never crossed 0.46 on this split under any protocol: 0.456 to 0.464 for the 010 checkpoints under the cosine schedule, 0.446 for the 025 replication (job 1598), 0.452 for the joint-fitness arm (Delta 22030924), and 0.459 for the best single run here ($\lambda = 10^{-1}$, seed 1, epoch 23).

Table 5: Each arm minus the no-penalty arm sharing its representation and budget, paired by seed, on the three readings. The fixed and minimum-loss readings pair complete runs; the max reading pairs every run (a partial run's max is a lower bound). t is the paired t statistic on $n - 1$ degrees of freedom, p two-sided; an arrow marks a difference with the same sign on every seed. Generated by `graph_reg_sweep_readout.py`.

arm	reading	n	difference by seed	mean	paired t	p
KL $\lambda = 10^{-5}$	epoch 29	3	-0.003, +0.002, -0.007	-0.0026	-0.9	0.448
KL $\lambda = 10^{-4}$	epoch 29	3	-0.005, +0.005, -0.016	-0.0054	-0.9	0.459
KL $\lambda = 10^{-3}$ (control)	epoch 29	3	+0.005, +0.013, +0.005	+0.0077 ↑	2.9	0.103
KL $\lambda = 10^{-2}$	epoch 29	3	+0.011, +0.014, +0.013	+0.0129 ↑	17.7	0.003
KL $\lambda = 10^{-1}$	epoch 29	3	+0.029, +0.023, +0.010	+0.0202 ↑	3.6	0.068
KL $\lambda = 1$	epoch 29	3	+0.021, +0.023, +0.021	+0.0218 ↑	30.0	0.001

arm	reading	n	difference by seed	mean	paired t	p
hard mask, layer 1	epoch 29	3	-0.001, -0.007, -0.003	-0.0036 ↓	-2.2	0.160
KL $\lambda = 10^{-3}$, random graphs	epoch 29	3	-0.007, +0.008, -0.015	-0.0047	-0.7	0.557
KL $\lambda = 10$	epoch 29	3	+0.029, +0.017, +0.013	+0.0195 ↑	4.0	0.056
KL $\lambda = 100$	epoch 29	3	+0.023, +0.016, +0.015	+0.0180 ↑	7.1	0.019
KL $\lambda = 10^{-1}$, random graphs	epoch 29	3	+0.010, +0.005, -0.005	+0.0034	0.8	0.518
KL $\lambda = 1$, random graphs	epoch 29	3	+0.008, +0.005, +0.001	+0.0045 ↑	2.2	0.160
hard mask, directed	epoch 29	3	-0.002, -0.001, -0.023	-0.0088 ↓	-1.2	0.352
hard mask, 2 hops, directed	epoch 29	3	-0.009, +0.001, -0.008	-0.0053	-1.6	0.252
hard mask, 3 hops, directed	epoch 29	3	-0.011, -0.005, -0.007	-0.0076 ↓	-4.3	0.050
KL $\lambda = 1$, 2-hop target	epoch 29	3	+0.008, +0.018, +0.014	+0.0131 ↑	4.3	0.051
KL $\lambda = 1$, 3-hop target	epoch 29	3	+0.010, +0.012, -0.010	+0.0041	0.6	0.619
KL $\lambda = 1$, symmetric target	epoch 29	3	+0.022, +0.019, +0.002	+0.0144 ↑	2.3	0.148
hard mask, layers 1–2	epoch 29	3	-0.002, -0.010, -0.006	-0.0061 ↓	-2.6	0.125
hard mask, layers 3–4	epoch 29	3	-0.008, -0.002, -0.018	-0.0093 ↓	-2.1	0.169
hard mask, layers 1–4	epoch 29	3	+0.004, +0.010, -0.009	+0.0016	0.3	0.805
KL $\lambda = 1$, layers 1–2	epoch 29	3	+0.016, +0.001, +0.018	+0.0120 ↑	2.2	0.154
KL $\lambda = 1$, layers 3–4	epoch 29	3	+0.018, +0.029, +0.015	+0.0206 ↑	4.9	0.039
hard mask, 60 epochs	epoch 29	3	+0.006, +0.009, -0.003	+0.0041	1.1	0.373
KL $\lambda = 1$, 60 epochs	epoch 29	3	+0.024, +0.021, +0.034	+0.0260 ↑	6.7	0.022
KL $\lambda = 1$, composite embedding	epoch 29	2	+0.423, +0.431	+0.4267 ↑	106.8	0.006
KL $\lambda = 10^{-5}$	max over epochs	3	-0.002, +0.002, -0.007	-0.0023	-0.9	0.455
KL $\lambda = 10^{-4}$	max over epochs	3	-0.000, +0.003, -0.006	-0.0013	-0.5	0.659
KL $\lambda = 10^{-3}$ (control)	max over epochs	3	+0.002, +0.001, +0.006	+0.0029 ↑	2.2	0.163
KL $\lambda = 10^{-2}$	max over epochs	3	+0.005, +0.005, +0.005	+0.0049 ↑	81.1	0.000
KL $\lambda = 10^{-1}$	max over epochs	3	+0.011, +0.008, -0.001	+0.0060	1.6	0.260
KL $\lambda = 1$	max over epochs	3	+0.007, +0.003, +0.003	+0.0041 ↑	3.0	0.094
hard mask, layer 1	max over epochs	3	-0.005, +0.003, -0.003	-0.0017	-0.7	0.557
KL $\lambda = 10^{-3}$, random graphs	max over epochs	3	-0.002, +0.004, -0.004	-0.0005	-0.2	0.857
KL $\lambda = 10$	max over epochs	3	+0.009, +0.003, +0.003	+0.0046 ↑	2.3	0.147
KL $\lambda = 100$	max over epochs	3	+0.010, -0.005, +0.004	+0.0031	0.7	0.570
KL $\lambda = 10^{-1}$, random graphs	max over epochs	3	-0.013, -0.008, -0.015	-0.0123 ↓	-6.1	0.026
KL $\lambda = 1$, random graphs	max over epochs	3	-0.011, -0.012, -0.009	-0.0106 ↓	-12.2	0.007
hard mask, directed	max over epochs	3	+0.001, +0.003, -0.002	+0.0006	0.5	0.673
hard mask, 2 hops, directed	max over epochs	3	-0.006, -0.001, -0.010	-0.0054 ↓	-2.1	0.174
hard mask, 3 hops, directed	max over epochs	3	-0.003, -0.004, -0.002	-0.0029 ↓	-3.6	0.069
KL $\lambda = 1$, 2-hop target	max over epochs	3	-0.010, -0.003, -0.003	-0.0052 ↓	-2.2	0.157
KL $\lambda = 1$, 3-hop target	max over epochs	3	+0.001, -0.008, -0.010	-0.0057	-1.7	0.229
KL $\lambda = 1$, symmetric target	max over epochs	3	-0.001, +0.006, +0.000	+0.0016	0.8	0.518
hard mask, layers 1–2	max over epochs	3	-0.001, -0.001, -0.003	-0.0016 ↓	-1.9	0.196
hard mask, layers 3–4	max over epochs	3	-0.002, -0.002, -0.003	-0.0024 ↓	-6.3	0.024
hard mask, layers 1–4	max over epochs	3	+0.000, +0.002, -0.005	-0.0012	-0.6	0.623
KL $\lambda = 1$, layers 1–2	max over epochs	3	+0.000, -0.006, +0.002	-0.0015	-0.6	0.610
KL $\lambda = 1$, layers 3–4	max over epochs	3	-0.002, +0.008, -0.002	+0.0012	0.4	0.755
hard mask, 60 epochs	max over epochs	3	+0.004, +0.000, +0.005	+0.0031 ↑	2.1	0.166
KL $\lambda = 1$, 60 epochs	max over epochs	3	+0.016, -0.001, +0.007	+0.0074	1.5	0.266
KL $\lambda = 1$, composite embedding	max over epochs	3	+0.409, +0.343, +0.300	+0.3506 ↑	11.1	0.008
KL $\lambda = 10^{-5}$	at min val loss	3	+0.001, +0.009, -0.004	+0.0020	0.5	0.657
KL $\lambda = 10^{-4}$	at min val loss	3	-0.002, +0.005, +0.000	+0.0012	0.6	0.635
KL $\lambda = 10^{-3}$ (control)	at min val loss	3	-0.002, +0.009, +0.004	+0.0037	1.1	0.380
KL $\lambda = 10^{-2}$	at min val loss	3	-0.002, +0.016, +0.006	+0.0066	1.3	0.335
KL $\lambda = 10^{-1}$	at min val loss	3	+0.015, +0.017, +0.005	+0.0124 ↑	3.3	0.080
KL $\lambda = 1$	at min val loss	3	+0.009, +0.005, -0.007	+0.0024	0.5	0.672
hard mask, layer 1	at min val loss	3	-0.005, +0.009, +0.001	+0.0018	0.4	0.714
KL $\lambda = 10^{-3}$, random graphs	at min val loss	3	-0.005, +0.015, +0.000	+0.0031	0.5	0.656
KL $\lambda = 10$	at min val loss	3	+0.004, +0.004, -0.003	+0.0018	0.8	0.494
KL $\lambda = 100$	at min val loss	3	+0.000, +0.002, -0.004	-0.0008	-0.5	0.679
KL $\lambda = 10^{-1}$, random graphs	at min val loss	3	-0.013, +0.003, -0.009	-0.0064	-1.4	0.309
KL $\lambda = 1$, random graphs	at min val loss	3	-0.018, -0.001, -0.010	-0.0096 ↓	-1.9	0.194
hard mask, directed	at min val loss	3	-0.003, +0.007, +0.002	+0.0021	0.8	0.523
hard mask, 2 hops, directed	at min val loss	3	-0.002, +0.005, -0.003	-0.0002	-0.1	0.949
hard mask, 3 hops, directed	at min val loss	3	-0.002, -0.001, -0.003	-0.0018 ↓	-2.9	0.103
KL $\lambda = 1$, 2-hop target	at min val loss	3	-0.009, -0.001, -0.001	-0.0037 ↓	-1.3	0.310
KL $\lambda = 1$, 3-hop target	at min val loss	3	-0.006, -0.005, -0.010	-0.0069 ↓	-4.6	0.044
KL $\lambda = 1$, symmetric target	at min val loss	3	-0.005, +0.017, +0.007	+0.0060	1.0	0.441
hard mask, layers 1–2	at min val loss	3	-0.002, +0.009, -0.007	-0.0001	-0.0	0.985
hard mask, layers 3–4	at min val loss	3	-0.001, +0.009, -0.002	+0.0020	0.6	0.632
hard mask, layers 1–4	at min val loss	3	+0.004, +0.012, -0.001	+0.0049	1.4	0.309
KL $\lambda = 1$, layers 1–2	at min val loss	3	-0.005, -0.002, +0.005	-0.0008	-0.3	0.816
KL $\lambda = 1$, layers 3–4	at min val loss	3	-0.007, +0.019, -0.010	+0.0007	0.1	0.948
hard mask, 60 epochs	at min val loss	3	-0.000, -0.003, +0.004	+0.0004	0.2	0.855

arm	reading	n	difference by seed	mean	paired t	p
KL $\lambda = 1$, 60 epochs	at min val loss	3	+0.016, +0.001, -0.003	+0.0048	0.8	0.487
KL $\lambda = 1$, composite embedding	at min val loss	2	+0.420, +0.422	+0.4210 ↑	714.4	0.001

3 What the prior does to training ✗

Support contraction happens at every λ, and saturates early. Without a penalty, layer-1 attention has nothing to do with the graphs: edge recall at degree is 0.006 at every diagnostic epoch, the value of attention unrelated to the adjacency (Figure 2c). The smallest weight tested already moves it to 0.782 at epoch 29 (0.759 at epoch 9), 10^{-4} to 0.866, 10^{-3} to 0.906, and from 10^{-2} up it sits at 0.914 to 0.915 with no further change to $\lambda = 1$. The divergence tells the same story with more resolution (Figure 2d): from 1,540 at 10^{-5} through 571 and 337 to about 300 at 10^{-2}, 10^{-1} and 1 (300, 297, 298), a floor the penalty cannot push below at any weight tested. Both diagnostics are flat from 10^{-2} to 1 while the fixed-epoch accuracy still rises from 0.438 to 0.447. How close the attention sits to the graph is set by $\lambda = 10^{-2}$; what the two larger weights add is not more contraction.

The gradient budget. The gradient probe (Figure 2e) measures the norm of each loss term's gradient on one batch before clipping. The point-loss gradient at epoch 0 is a property of the seed, not the arm: 7.4, 32.3 and 2.8 for seeds 1, 2 and 3 in every arm that shares the initialization (the mask arm, with a different layer-1 kernel, reads 5.7, 32.9 and 2.7). The penalty's gradient scales with λ as it must, 1.1 at 10^{-3}, 10.9 at 10^{-2}, 109 at 10^{-1} and 1,097 at 1 in the seed mean at epoch 0, and grows over training while the point-loss gradient settles to 1.0 to 4.3: at epoch 20 the ratio is 1.1 to 1.5 at 10^{-3}, 15 to 24 at 10^{-2}, 86 to 146 at 10^{-1} and 680 to 1,550 at $\lambda = 1$. The two best arms of the ladder train under a total gradient that is almost entirely the penalty. The trainer clips the total gradient norm at 10, so at $\lambda \geq 10^{-1}$ every update is rescaled by two to three orders of magnitude before the optimizer sees it.

Hypothesis (untested): AdamW normalizes each parameter's update by its running gradient scale, so a uniform rescaling of the whole gradient leaves the update nearly unchanged, and the penalty's gradient is concentrated in the parameters that shape layer-1 attention (the layer-1 query and key projections and, through them, the gene table). Under that reading the arms at $\lambda \geq 10^{-1}$ train the rest of the network on the point loss much as the no-penalty arm does, which is consistent with their training Pearson matching it (0.625 and 0.624 against 0.624), while layer 1 is held on the graph for the whole run. The mechanism is not measured here; a per-parameter-group norm on the probe batch would measure it.

The prior is not a fit-reducing regularizer at the weights that help. Training Pearson at epoch 29 (Figure 2h) is 0.624 with no penalty, 0.636 at 10^{-3}, 0.613 at 10^{-2}, 0.625 at 10^{-1} and 0.624 at 1, so the arms that generalize best at epoch 29 fit the training triples as well as the arm that generalizes worst. The two smallest weights fit the training set better than any other arm, 0.657 at 10^{-5} and 0.663 at 10^{-4}, and generalize no better than no penalty. The validation point loss (Figure 2i) reaches its minimum near epoch 13 in every arm (0.80 to 0.82) and then rises; at epoch 29 it is 0.867 with no penalty and 0.869 under the mask against 0.826 and 0.825 at $\lambda = 10^{-1}$ and 1. The prior at high weight does not stop the model from fitting the training triples; it slows the growth of held-out error after the point-loss minimum.

Hypothesis (untested): the late decline of the unpenalized model is layer-1 attention drifting away from any fixed structure as the gene table specializes to the training triples, and a prior strong enough to hold layer 1 in place removes that degree of freedom without touching the rest. The mask removes the same freedom in a different way, by fixing the support and leaving the weights free, and it declines exactly as no penalty does, so if this hypothesis is right the part of layer 1 that drifts is the weighting within the support, not the support.

The whole objective, term by term. The loss the optimizer steps on is the point loss plus the distribution term, the Sinkhorn-regularized Wasserstein distance between the batch's predicted and observed label distributions at weight 0.1, plus the penalty of Eq. 1. Figure 3 draws the three terms and their sum by epoch, on the training steps and on the validation pass, for the arms of Figure 2g to i and $\lambda = 100$. The sum is the penalty from $\lambda = 10^{-1}$ up: at epoch 29 the training total is 0.62 with no penalty and 0.64 under the mask against 0.94 at 10^{-3}, 30.6 at 10^{-1}, 299 at 1 and 30,100 at 100, the point loss underneath being 0.59 to 0.66 in every arm. The distribution term is small and the same everywhere: 0.032 at epoch 0 and 0.010 to 0.012 at epoch 29 on the training steps (3 percent of the point loss, then 2), 0.029 to 0.032 and 0.019 to 0.022 on the validation pass; it is not varied here and does not separate the arms. The penalty starts high and floors fast: $\lambda\times$ 16,600 on the first training epoch and $\lambda\times$ 2,150 on the first validation pass, down to $\lambda\times$ 360 by epoch 13 and $\lambda\times$ 300 at epoch 29, the same floor on training and validation, which is the divergence of Figure 2d read along the run. The random-graph arm at $\lambda = 1$ floors three times higher, at 910, because a degree-matched rewiring is a harder target for the attention to match than the stored graph.

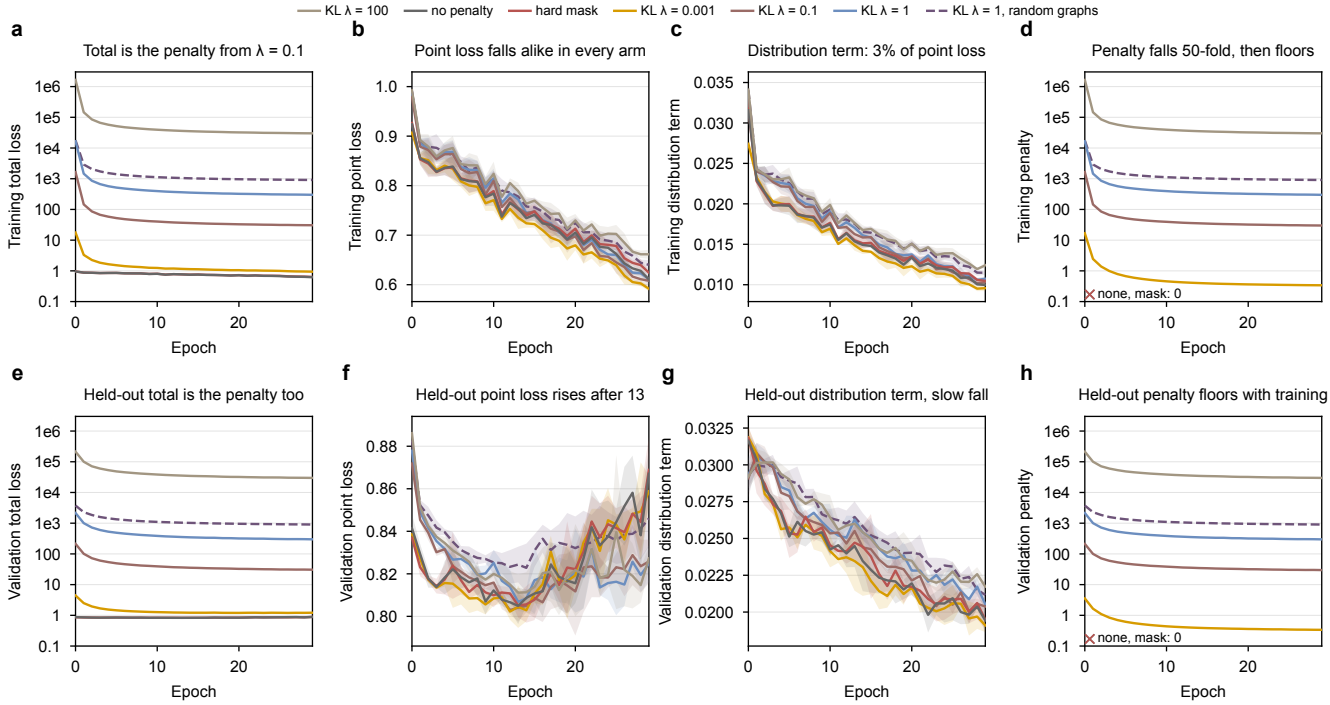


Figure 3: The training objective term by term, mean over complete seeds with a ± sd band, for no penalty, the hard mask, the prior at $\lambda = 10^{-3}$, 10^{-1}, 1 and 100, and the prior at $\lambda = 1$ toward random graphs (dashed); read from W&B by `experiments/025-solid-growth/scripts/graph_reg_sweep_readout.py` and drawn by `experiments/025-solid-growth/scripts/graph_reg_sweep_plots.py`. Top row, the training steps averaged over each epoch; bottom row, the validation pass at the end of each epoch. **a**, **e**, The total the optimizer steps on, point loss plus 0.1 times the Wasserstein distance plus the penalty, log scale. **b**, **f**, The point loss, the z-scored mean squared error. **c**, **g**, The weighted distribution term, 0.1 times the Sinkhorn Wasserstein distance between the batch's predicted and observed label distributions. **d**, **h**, The penalty, λ times the divergence, log scale; the two arms without a penalty are at 0.

4 The random-graph control: biology or any target ✗

The control arm keeps everything of the biological arm at the same λ except the graphs: each of the nine is rewired before the model sees it, by double-edge swaps that keep every gene's degree, five swaps per edge, seeded by the run seed (`cgt_s0_r_kl_rand_031`, `torchcell.graph.rewire`). The prior pulls the heads toward the rewired graphs and the edge diagnostics are scored against them. The arm ran first at 10^{-3}, the 010 operating point, with three seeds, and in round 1b at the two λ where the biological prior's gain is largest, 10^{-1} and 1, with five seeds each (Delta jobs 22461957 to 22461967). The paired comparisons below use seeds 1 to 3, the seeds the biological arms have.

The heads learn the random graphs, less completely. Edge recall at degree against the rewired graphs reaches 0.814 at 10^{-3} and 0.822 at 10^{-1} and 1, with divergence 980, 902 and 904 (Figure 2c, d; Table 4), against 0.906 to 0.915 and 337 to 298 for the biological graphs at the same λ. A random graph of the same degree sequence is a target the attention can and does move toward, but it does not get as close in 30 epochs at any λ, and raising λ a hundredfold closes none of the gap. Whether that is because the biological graphs are more compatible with the point loss, or because the rewired graphs are less clustered and so harder to represent with a low-rank attention, is not separated by anything logged.

At $\lambda = 10^{-3}$ the accuracy question is underpowered. On the fixed-epoch reading the three random-graph seeds give 0.417, 0.433 and 0.410, against 0.429, 0.438 and 0.430 for the biological graphs on the same seeds and 0.424, 0.425 and 0.425 with no penalty. The biological graphs are above the random ones on every seed, by 0.012, 0.005 and 0.020 (mean +0.012, paired $t = 2.8$, $p = 0.11$); the random ones straddle no penalty, with paired differences of −0.008, +0.008 and −0.015 (mean −0.005, $t = -0.7$, $p = 0.56$). At the max-over-epochs reading the three arms are 0.451, 0.447 and 0.448 (biological minus random $p = 0.44$), and at minimum validation loss 0.444, 0.444 and 0.440 ($p = 0.86$). The effect being tested at this λ is the 0.005 to 0.013 that the biological prior adds over no penalty at epoch 29, and three seeds of a control whose own seed spread is 0.012 cannot resolve it.

At $\lambda = 10^{-1}$ and 1 the biology separates from the conditioning. The five random-graph seeds give 0.430 ± 0.006 at 10^{-1} and 0.427 ± 0.004 at $\lambda = 1$ on the fixed-epoch reading, 0.437 ± 0.003 and 0.435 ± 0.004 at the max over epochs, and 0.434 ± 0.004 and 0.430 ± 0.006 at minimum validation loss (Table 3). Paired on seeds 1 to 3, the biological graphs are above the random ones on every seed under every reading (Figure 2f). At 10^{-1} the differences are +0.019, +0.017

and +0.015 at epoch 29 (mean +0.017, $t = 14.5$, $p = 0.005$), +0.025, +0.017 and +0.014 at the max (+0.018, $t = 5.6$, $p = 0.03$) and +0.029, +0.014 and +0.014 at minimum loss (+0.019, $t = 3.9$, $p = 0.06$). At $\lambda = 1$ they are +0.014, +0.018 and +0.020 at epoch 29 (+0.017, $t = 8.8$, $p = 0.013$), +0.018, +0.015 and +0.012 at the max (+0.015, $t = 8.3$, $p = 0.014$) and +0.027, +0.006 and +0.003 at minimum loss (+0.012, $t = 1.6$, $p = 0.26$). Against no penalty the random arms are not the biological arms: at epoch 29 they sit at or just above it (+0.003 at 10^{-1}, $p = 0.52$; +0.004 at 1, up on every seed, $p = 0.16$), where the biological prior is +0.020 and +0.022 above; at the max over epochs they are below it on every seed, by −0.012 at 10^{-1} ($t = -6.1$, $p = 0.026$) and −0.011 at 1 ($t = -12.2$, $p = 0.007$), with a minimum validation point loss of 0.813 and 0.816 against 0.801 (Table 4). A strong prior toward degree-matched random graphs therefore removes little of the late decline and lowers the peak; the fixed-epoch gain of the biological prior is not reproduced by conditioning on any graph of the same degree sequence. The gain over random at the peak reading, 0.015 to 0.018, is larger than the biological arm's own gain over no penalty there (0.004 to 0.006), because most of it is the random arm falling below no penalty rather than the biological arm rising above it. What the biological graphs buy a best-checkpoint deployment remains at most 0.006; what they buy a fixed-budget run is 0.02, and that 0.02 is theirs.

5 Reading, and what is still missing ✗

On the hypothesis. Stated as proposed: the graphs help, but only when the model is optimized through the divergence; imposing them as a mask does not help. The mask half holds on both readings and all three seeds: the mask is indistinguishable from no penalty (0.421 against 0.425 at epoch 29, 0.446 against 0.448 at the best epoch), and it trains without the collapse the earlier cosine-schedule runs showed. The prior half holds on the fixed-epoch reading from $\lambda = 10^{-3}$ up, with the gain growing to 0.021 to 0.023 at $\lambda = 1$ and holding through $\lambda = 100$, and holds weakly on the best-epoch reading (0.003 to 0.007 at $\lambda = 1$, positive on every seed). The word "help" needs its reading attached: a best-checkpoint deployment gains at most 0.006 from the prior; a fixed-budget run gains 0.02, because the prior removes the decline after epoch 16 rather than raising the peak. The random-graph control at $\lambda = 10^{-1}$ and 1 (Section 4) establishes that the help comes from the biology of the graphs: a prior of the same weight toward degree-matched rewirings gives 0.003 to 0.004 over no penalty at epoch 29 where the biological graphs give 0.020 to 0.022, and at the best epoch it falls 0.011 to 0.012 below no penalty on every seed. The alternative, that a strong prior toward any graph of the same degree sequence helps as much, is rejected at both λ; what stays open is whether the biological gain at the peak (0.004 to 0.006 over no penalty) is worth a prior that costs 0.011 when its target is wrong.

What the mock-up predicted and what was measured. The figure design (`notes/assets/drawio/FigS-graph-regularization-sweep.gen.py`) drew six panels. Panel a, contraction with λ: measured, and it saturates at 10^{-2}. Panel b, accuracy peaking between the extremes: measured, and there is no interior peak; the maximum is at the top of the ladder. Panel c, the penalty carrying the gradient before the point loss can: measured, and the penalty dominates the gradient at every probe epoch for $\lambda \geq 10^{-2}$, not only early. Panels d and e, discovery of edges outside a head's prior and overruling of cataloged edges by the digenic interaction, need the layer-1 attention of the saved checkpoints, which are on Delta `/scratch` and have not been read; the edge-recovery precision logged at $k = 32$ (0.21 for every prior arm from 10^{-3} up, 0.006 with no penalty) is the nearest logged quantity and is not the discovery statistic. Panel f, the random-graph control: measured first at the wrong λ, then in round 1b at the two λ where the gain is largest, where it separates.

The panel titles, checked. Each title of Figure 2 states a finding, which is a place for bias to enter, so each is set against the number it rests on. (a) "Prior gains at epoch 29, little at the peak": +0.022 at $\lambda = 1$ on the fixed reading against +0.004 on the max reading (Table 5). (b) "Held-out loss falls with λ from 0.001 up": 0.859, 0.839, 0.826, 0.825 at 10^{-3} to 1 against 0.867 with no penalty; the two smallest weights are above no penalty (0.880, 0.878), which the title's "from 0.001 up" carries (Table 3). (c) "Any λ tested moves attention onto its target": 0.782 at 10^{-5} against 0.006 at 0. (d) "Divergence floors near 300 from $\lambda = 0.01$": 300, 297, 298, 298, 297, 298 from 10^{-2} to 100. (e) "Penalty dominates the gradient from $\lambda = 0.01$": ratio 15 to 24 at 10^{-2} by epoch 20 against 1.1 to 1.5 at 10^{-3}, and 1.0×10^4 and 9.6×10^4 at 10 and 100. (f) "Biology beats random graphs by ≈0.017 at $\lambda \geq 0.1$": biological minus random on seeds 1 to 3 is +0.017 at epoch 29 at both 10^{-1} ($t = 14.5$, $p = 0.005$) and 1 ($t = 8.8$, $p = 0.013$), +0.018 and +0.015 at the max ($p = 0.03$ and 0.014), and +0.019 and +0.012 at minimum validation loss ($p = 0.06$ and 0.26); at 10^{-3} it is +0.012 at epoch 29 ($p = 0.11$) and within noise on the other readings. (g) "Strong prior removes the decline after epoch 16": seed-mean 0.441 to 0.425 with no penalty against 0.435 to 0.447 at $\lambda = 1$ and 0.441 to 0.445 at 10^{-1}. (h) "Strong priors fit training like no penalty": 0.625 and 0.624 against 0.624. (i) "Held-out loss rises less under a strong prior": from the epoch-13 minimum, 0.801 to 0.867 with no penalty against 0.803 to 0.825 at $\lambda = 1$.

Open items.

1. Closed in round 1b: $\lambda = 10$ and 100 (Delta jobs 22461968 to 22461973) found no turnover (Section 2), and seed 2 of $\lambda = 10^{-5}$, whose first job (22055161) never reached W&B, is in the tables.

2. The mechanism and budget figures the round-2 wireframe promised (Figure 4, rows 2 and 3) are not drawn; their numbers are in Tables 3, 5 and 6 and read in Section 6.

3. Panels d and e from the checkpoints: pull the best-Pearson checkpoint of one seed per arm from Delta and score layer-1 attention against TFLink-only and STRING-experimental-only edges and against Costanzo digenic ε.

4. A per-parameter-group gradient probe, to test the hypothesis of Section 3 about where the penalty's gradient lands.

Table 6: Every run read, one row per seed. Epochs is the number of validation epochs logged, with the count of epochs whose validation Pearson was undefined (NaN, constant predictions) where there were any; Pearson at epoch 29 and the max over epochs with its epoch; the gradient-probe ratio (penalty over point loss) at epoch 0; the W&B run id in project `torchcell_025-solid-growth_equivariant_cell_graph_transformer`. Generated by `graph_reg_sweep_readout.py`.

arm	seed	epochs	Pearson, ep 29	max (epoch)	gradient ratio, ep 0	W&B run
no penalty ($\lambda = 0$)	1	30	0.4243	0.4476 (16)	0.00	`zvtj3waw`
no penalty ($\lambda = 0$)	2	30	0.4248	0.4460 (15)	0.00	`4vgeyw2i`
no penalty ($\lambda = 0$)	3	30	0.4253	0.4493 (16)	0.00	`mmiydyws`
KL $\lambda = 10^{-5}$	1	30	0.4213	0.4456 (13)	0.00	`rldsrcnr`
KL $\lambda = 10^{-5}$	2	30	0.4272	0.4479 (15)	0.00	`8ttjk7ru`
KL $\lambda = 10^{-5}$	3	30	0.4180	0.4425 (14)	0.00	`7fx9njte`
KL $\lambda = 10^{-4}$	1	30	0.4189	0.4473 (16)	0.01	`r2i3i9xq`
KL $\lambda = 10^{-4}$	2	30	0.4297	0.4486 (15)	0.00	`h8zhfq2d`
KL $\lambda = 10^{-4}$	3	30	0.4096	0.4431 (6)	0.04	`oxh1ug5l`
KL $\lambda = 10^{-3}$ (control)	1	30	0.4292	0.4498 (16)	0.14	`6dvbzy8w`
KL $\lambda = 10^{-3}$ (control)	2	30	0.4378	0.4471 (12)	0.04	`ppm9kl4w`
KL $\lambda = 10^{-3}$ (control)	3	30	0.4304	0.4548 (14)	0.40	`22u65gax`
KL $\lambda = 10^{-2}$	1	30	0.4358	0.4524 (27)	1.45	`u1hdm0y0`
KL $\lambda = 10^{-2}$	2	30	0.4386	0.4510 (19)	0.37	`eufg8mlm`
KL $\lambda = 10^{-2}$	3	30	0.4386	0.4541 (14)	4.01	`0vgat9et`
KL $\lambda = 10^{-1}$	1	30	0.4528	0.4590 (23)	14.47	`lz1rd90p`
KL $\lambda = 10^{-1}$	2	30	0.4473	0.4542 (23)	3.69	`df8jt886`
KL $\lambda = 10^{-1}$	3	30	0.4349	0.4478 (13)	40.20	`5pz6vc5y`
KL $\lambda = 1$	1	30	0.4456	0.4545 (26)	146.51	`367oc8pv`
KL $\lambda = 1$	2	30	0.4480	0.4487 (23)	37.17	`ds928dmp`
KL $\lambda = 1$	3	30	0.4461	0.4521 (24)	399.95	`zxme1g1p`
hard mask, layer 1	1	30	0.4232	0.4430 (18)	0.00	`g35yf77n`
hard mask, layer 1	2	30	0.4181	0.4491 (15)	0.00	`1pv2lpf9`
hard mask, layer 1	3	30	0.4222	0.4458 (17)	0.00	`tpixauk7`
KL $\lambda = 10^{-3}$, random graphs	1	30	0.4169	0.4459 (13)	0.15	`usgha5w6`
KL $\lambda = 10^{-3}$, random graphs	2	30	0.4329	0.4500 (19)	0.04	`9tqgl66k`
KL $\lambda = 10^{-3}$, random graphs	3	30	0.4104	0.4456 (14)	0.41	`zuw5z7k4`
KL $\lambda = 10$	1	30	0.4532	0.4562 (24)	1457.92	`zs8hylbw`
KL $\lambda = 10$	2	30	0.4415	0.4486 (17)	369.90	`tkp6zlre`
KL $\lambda = 10$	3	30	0.4381	0.4519 (24)	3988.77	`j8xlrvrb`
KL $\lambda = 100$	1	30	0.4474	0.4580 (27)	14960.61	`hx4s7bxa`
KL $\lambda = 100$	2	30	0.4406	0.4406 (29)	3783.01	`i1l1jhm8`
KL $\lambda = 100$	3	30	0.4403	0.4535 (24)	40438.52	`dnlc6pib`
KL $\lambda = 10^{-1}$, random graphs	1	30	0.4343	0.4343 (29)	14.78	`u6he7prk`
KL $\lambda = 10^{-1}$, random graphs	2	30	0.4300	0.4376 (12)	3.71	`h7wumv3p`
KL $\lambda = 10^{-1}$, random graphs	3	30	0.4203	0.4341 (19)	40.87	`w7wt6bmo`
KL $\lambda = 10^{-1}$, random graphs	4	30	0.4294	0.4412 (21)	2.14	`7e9caxfo`
KL $\lambda = 10^{-1}$, random graphs	5	30	0.4372	0.4374 (24)	22.20	`jpmuaaah`
KL $\lambda = 1$, random graphs	1	30	0.4321	0.4366 (28)	149.41	`l6hv7cub`
KL $\lambda = 1$, random graphs	2	30	0.4298	0.4342 (14)	37.41	`awnp9vvt`
KL $\lambda = 1$, random graphs	3	30	0.4260	0.4404 (22)	405.90	`gtmbi3fw`
KL $\lambda = 1$, random graphs	4	30	0.4256	0.4348 (28)	20.97	`urwi2lxu`
KL $\lambda = 1$, random graphs	5	30	0.4212	0.4287 (19)	223.91	`n0vst095`
hard mask, directed	1	30	0.4228	0.4482 (19)	0.00	`ojdk2wjb`
hard mask, directed	2	30	0.4233	0.4490 (17)	0.00	`pl7w8p23`
hard mask, directed	3	30	0.4020	0.4477 (14)	0.00	`ofajio46`
hard mask, 2 hops, directed	1	30	0.4151	0.4419 (13)	0.00	`5tnf7v0s`
hard mask, 2 hops, directed	2	30	0.4261	0.4453 (15)	0.00	`ol8loxg2`
hard mask, 2 hops, directed	3	30	0.4172	0.4395 (13)	0.00	`z3yp3qvg`
hard mask, 3 hops, directed	1	30	0.4135	0.4448 (18)	0.00	`aypj6d9b`
hard mask, 3 hops, directed	2	30	0.4200	0.4417 (15)	0.00	`6qzsjdj2`
hard mask, 3 hops, directed	3	30	0.4180	0.4477 (14)	0.00	`n4r2coej`
KL $\lambda = 1$, 2-hop target	1	30	0.4319	0.4376 (25)	67.88	`ys7mz0e8`
KL $\lambda = 1$, 2-hop target	2	30	0.4430	0.4432 (28)	18.82	`p7us24x7`
KL $\lambda = 1$, 2-hop target	3	30	0.4390	0.4463 (22)	181.67	`qhypj06y`

arm	seed	epochs	Pearson, ep 29	max (epoch)	gradient ratio, ep 0	W&B run
KL $\lambda = 1$, 3-hop target	1	30	0.4343	0.4484 (28)	43.67	`tly5od5x`
KL $\lambda = 1$, 3-hop target	2	30	0.4368	0.4382 (26)	12.36	`b8joneh6`
KL $\lambda = 1$, 3-hop target	3	30	0.4155	0.4392 (28)	113.27	`95fs8pfw`
KL $\lambda = 1$, symmetric target	1	30	0.4465	0.4465 (29)	114.54	`3faippxa`
KL $\lambda = 1$, symmetric target	2	30	0.4437	0.4515 (27)	28.32	`2rb2lshh`
KL $\lambda = 1$, symmetric target	3	30	0.4273	0.4496 (22)	333.25	`16l3gkau`
hard mask, layers 1–2	1	30	0.4222	0.4468 (16)	0.00	`jcneeoxl`
hard mask, layers 1–2	2	30	0.4144	0.4453 (15)	0.00	`do0jnbry`
hard mask, layers 1–2	3	30	0.4196	0.4460 (14)	0.00	`677afro6`
hard mask, layers 3–4	1	30	0.4163	0.4453 (13)	0.00	`udcsvf86`
hard mask, layers 3–4	2	30	0.4224	0.4442 (12)	0.00	`u2cqgyt0`
hard mask, layers 3–4	3	30	0.4078	0.4461 (14)	0.00	`trha5wbm`
hard mask, layers 1–4	1	30	0.4279	0.4478 (18)	0.00	`xpz6qsw7`
hard mask, layers 1–4	2	30	0.4348	0.4475 (15)	0.00	`hov78sie`
hard mask, layers 1–4	3	30	0.4164	0.4440 (14)	0.00	`k8rhye1o`
KL $\lambda = 1$, layers 1–2	1	30	0.4407	0.4480 (27)	191.86	`4htu7x9c`
KL $\lambda = 1$, layers 1–2	2	30	0.4262	0.4397 (22)	54.02	`jvhj6w7p`
KL $\lambda = 1$, layers 1–2	3	30	0.4436	0.4508 (28)	581.12	`0py9yk6e`
KL $\lambda = 1$, layers 3–4	1	30	0.4421	0.4453 (26)	242.57	`zy2us0hw`
KL $\lambda = 1$, layers 3–4	2	30	0.4537	0.4537 (29)	56.02	`175ssbar`
KL $\lambda = 1$, layers 3–4	3	30	0.4405	0.4475 (25)	797.09	`hpm8im1i`
no penalty, 60 epochs	1	60	0.4140	0.4412 (15)	0.00	`htqyeonz`
no penalty, 60 epochs	2	60	0.4188	0.4477 (15)	0.00	`ye4iu9ra`
no penalty, 60 epochs	3	60	0.4133	0.4405 (14)	0.00	`beggtb67`
hard mask, 60 epochs	1	60	0.4201	0.4455 (18)	0.00	`680dsbq0`
hard mask, 60 epochs	2	60	0.4278	0.4479 (17)	0.00	`rol0rp8k`
hard mask, 60 epochs	3	60	0.4104	0.4451 (15)	0.00	`038gk8dm`
KL $\lambda = 1$, 60 epochs	1	60	0.4379	0.4571 (20)	146.51	`o3dyrd28`
KL $\lambda = 1$, 60 epochs	2	60	0.4393	0.4469 (23)	37.17	`fb3jeniw`
KL $\lambda = 1$, 60 epochs	3	60	0.4468	0.4475 (46)	399.95	`vdtfq7fb`
no penalty, composite embedding	1	30 (5 NaN)	-0.0084	0.0091 (12)	0.00	`3rswgmkm`
no penalty, composite embedding	2	30 (11 NaN)	–	0.0784 (15)	0.00	`ev4kv036`
no penalty, composite embedding	3	30 (6 NaN)	-0.0030	0.1303 (9)	0.00	`tma2rpjf`
KL $\lambda = 1$, composite embedding	1	30	0.4142	0.4179 (28)	49.33	`1aniekse`
KL $\lambda = 1$, composite embedding	2	30	0.4212	0.4212 (29)	546.91	`ng5kw1h9`
KL $\lambda = 1$, composite embedding	3	30	0.4276	0.4307 (28)	54.65	`583mvatn`

6 The next rounds: what finishes the figure, what decides the mechanism ✗

The first five subsections are the design of the follow-up Delta rounds, written before submission so the figures they feed are fixed in advance, as the first round's mock-up was (Figure 1); nothing in them is measured. The rounds ran between 2026-10-01 and 2026-10-10 and the last subsection reads them. Every arm is the same protocol as round 1, the control config with the named keys changed, three seeds, on Delta `gpuA40x4`, so every new column is read against the columns already in hand. Table 8 is generated from `experiments/025-solid-growth/scripts/graph_reg_round2_plan.py`, which is also where the launcher will read the arm list from once a round is approved. The rounds are ordered by the two questions the study asks. Round 1b completes Figure 2 and answers them where the effect is: is the prior better than the mask (the top of the ladder), and do the graphs help (the random control at the λ that separates). Round 2 asks the question that follows at once from "do the graphs help": how far along the graph should a head reach, two hops and three, for both mechanisms on identical targets, with direction beside it. Round 3 holds what is optional or orthogonal: where the graph block sits, the training budget, the representation, the width. The random-graph arm of round 1 ran the full 30-epoch protocol on all three seeds (30 logged epochs each), so its comparison in Figure 2f is at matched budget.

What round 1 left open, and the arm that closes each.

1. *Biology or conditioning.* The random-graph control ran at $\lambda = 10^{-3}$, where the biological prior adds 0.005 to 0.013 and three seeds do not resolve it (random against no penalty -0.005 ± 0.012, $p = 0.56$). At $\lambda = 1$ the biological prior adds 0.021 to 0.023 on every seed. Three seeds of the rewired graphs at $\lambda = 1$ decide whether that is the graphs or the penalty; three more at 10^{-1} cover the loss-selected optimum.

2. *The top of the ladder.* Accuracy at epoch 29 rises to $\lambda = 1$ with no turnover inside the ladder, while contraction and divergence are flat from 10^{-2}. $\lambda = 10$ and 100 find the turnover, or show there is none in reach.

3. *Direction.* The prior's target is directed, $A[\text{source}, \text{target}]$ row-normalized; the mask builder symmetrizes it. So the mask-versus-prior gap confounds soft-versus-hard with directed-versus-undirected on the regulatory and TFLink heads. Two arms separate them: the prior at $\lambda = 1$ toward symmetrized targets, and a directed mask (the transpose write removed, which leaves the seven symmetric graphs unchanged). If direction matters, the symmetric prior falls toward the mask and the directed mask rises toward the prior. A two-head variant, one head on A and one on A^{T} for each directed graph, is the follow-up if the directed mask helps, not part of this round.

4. *Reach.* A head under either mechanism sees one hop. The k-hop support is the indicator of $A \cup A^2 \cup \cdots \cup A^k$; the mask restricts attention to it (self-loops kept) and the prior's target is its row normalization with the diagonal of the one-hop target (the cell graph gives every gene a self-loop, so every round-1 target already held a self entry in every row), so the two mechanisms are compared on identical two- and three-hop targets and differ from round 1 in reach alone. Table 7 is the density of that support, measured on the graphs the trainer builds. It settles one design choice: with edges made undirected, two hops of the physical, co-expression or experimental graph already reach 60 to 70 percent of all ordered gene pairs and three hops 75 to 94 percent, so a symmetric two-hop mask is close to no mask on those heads. The reach arms therefore use the *directed* k-hop support for both mechanisms, which is what the prior has always used, and the directed one-hop mask is their one-hop anchor. Under the stored direction the regulatory and TFLink graphs stay sparse at three hops (1 and 3 percent of pairs); the seven undirected graphs are symmetric by construction, so on those heads the two- and three-hop arms measure how a nearly uniform target behaves, which is a finding about the graph rather than a reason to skip the arm.

5. *Placement (optional).* The encoder has eight layers and the graphs enter layer 1 only, an $[S, M, S, S, S, S, S, S]$ pattern. Round 3 widens the block and shifts it: layers 1 to 2 $[S, M, M, S, \ldots]$, layers 3 to 4 $[S, S, S, M, M, S, S, S]$, and layers 1 to 4 $[S, M, M, M, M, S, S, S]$, each for the mask and for the prior at $\lambda = 1$. The shifted block asks whether the graphs belong where genes are first mixed or after two free layers have formed a representation. The mask costs nothing in memory; each regularized layer of the prior materializes its attention, about 6 GB per layer, which is why the prior stops at four layers.

6. *Budget.* No penalty peaks at epoch 16 and declines; $\lambda = 1$ peaks at 23 to 26 and was still at its high at epoch 29. Sixty epochs of no penalty, the mask and $\lambda = 1$ show whether the strong prior keeps rising or only declines later, and whether the fixed-budget gap grows or closes. These need the 48 h clock.

7. *Completion.* Seed 2 of $\lambda = 10^{-5}$ never reached W&B (Delta job 22055161); one run fills it. The third random-graph seed at 10^{-3} finishes on its own.

8. *Representation.* Every arm here uses the learnable gene table, the representation that memorizes fastest, and the prior's gain is mostly resistance to the late decline. The composite sequence embedding (`embfit_035` without its fitness head) resisted that decline on its own: on this split with the fitness head it reads +0.016 over the table at epoch 29 and nothing at the peak. Composite with and without the prior at $\lambda = 1$, three seeds each, says whether the graph result survives the representation the model will actually use.

9. *Width.* The hidden width of 180 is inherited from 010 and was never swept on this task. Doubling it to 360 (heads of 40) takes the transformer from 3.1 M to about 12.5 M parameters and the model from 4.8 M to about 16.6 M; memory is set by the $9 \times 6{,}608^2$ attention of the regularized layer and does not move, the epoch should take 1.5 to 2.5 times as long, which is the 48 h clock. Table and composite, each with and without the prior, so width is read against the two representations and the prior at once; the composite preprocessor is re-matched to the table's parameter count at the new width.

Table 7: Reach of the nine gene-gene graphs, on the 6,607 genes the trainer uses: the mean number of genes reachable from a gene within one, two and three steps, and the fraction of all ordered pairs that is, along the stored direction and with every edge made undirected (the two coincide for the seven undirected graphs). Generated by `graph_reg_khop_density.py`.

		mean reach per gene			fraction of all pairs		
graph	edges	1 hop	2 hops	3 hops	1 hop	2 hops	3 hops
physical	undirected	42	4,057	4,942	0.006	0.614	0.748
regulatory	directed	6	27	69	0.001	0.004	0.010
regulatory	undirected	11	3,025	5,387	0.002	0.458	0.815
tflink	directed	30	170	208	0.005	0.026	0.032
tflink	undirected	60	3,869	3,896	0.009	0.586	0.590
string12_0_neighborhood	undirected	44	546	719	0.007	0.083	0.109
string12_0_fusion	undirected	4	36	226	0.001	0.006	0.034
string12_0_cooccurence	undirected	3	9	21	0.001	0.001	0.003
string12_0_coexpression	undirected	302	4,587	6,195	0.046	0.694	0.938
string12_0_experimental	undirected	249	4,457	5,430	0.038	0.675	0.822
string12_0_database	undirected	22	246	1,074	0.003	0.037	0.163

Table 8: The next rounds, in submission order. Every arm is `cgt_s0_r_kl_ctrl_013` with the change named, three seeds unless stated, 30 epochs on the 24 h clock or 60 on the 48 h clock. GPU-hours are four A40s times the measured 27 min per epoch times seeds. "code" names the model change an arm needs before it can be launched; the others are Hydra overrides of keys that exist. Round 1b finishes Figure 2; round 2 fills the mechanism figure; round 3 the budget and representation figure. Generated by `graph_reg_round2_plan.py`.

round	question	arm	change on `ctrl_013`	code	seeds	epochs	GPU-h
1b	do graphs help	random graphs, KL 1	`rand_031`, `graph_reg_lambda=1`	none	5	30	270
		random graphs, KL 0.1	`rand_031`, `graph_reg_lambda=0.1`	none	5	30	270
	prior vs mask	KL 10	`graph_reg_lambda=10`	none	3	30	162
		KL 100	`graph_reg_lambda=100`	none	3	30	162
	completion	KL 1e-5, seed 2	`graph_reg_lambda=1e-5`, `seed=2`	none	1	30	54
2	reach	mask, 2-hop, directed	`attention_mask hops=2`, `symmetric=false`	k-hop support: reachability within k steps, self-loops kept	3	30	162
		mask, 3-hop, directed	`attention_mask hops=3`, `symmetric=false`	same	3	30	162
		KL 1, 2-hop target	`graph_regularization hops=2`	k-hop target: row-normalized k-hop indicator	3	30	162
		KL 1, 3-hop target	`graph_regularization hops=3`	same	3	30	162
	direction	KL 1, symmetric targets	`graph_regularization symmetrize=true`	symmetrize A for the directed graphs before row normalization	3	30	162
		mask, directed	`attention_mask symmetric=false`	drop the transpose write in the mask builder	3	30	162
3	placement (optional)	mask, layers 1-2	`attention_mask layers=[1,2]`	none	3	30	162
		mask, layers 3-4	`attention_mask layers=[3,4]`	none	3	30	162
		mask, layers 1-4	`attention_mask layers=[1,2,3,4]`	none	3	30	162
		KL 1, layers 1-2	`graph_reg_lambda=1`, `graph_reg_layer=[1,2]`	none	3	30	162
		KL 1, layers 3-4	`graph_reg_lambda=1`, `graph_reg_layer=[3,4]`	none	3	30	162
		KL 1, layers 1-4	`graph_reg_lambda=1`, `graph_reg_layer=[1,2,3,4]`	none	3	30	162
	budget	no penalty, 60 epochs	`graph_reg_lambda=0`, `max_epochs=60`	none	3	60	324
		mask, 60 epochs	`mask_028`, `max_epochs=60`	none	3	60	324
		KL 1, 60 epochs	`graph_reg_lambda=1`, `max_epochs=60`	none	3	60	324
	representation	composite embedding, no penalty	`emb_040`, `graph_reg_lambda=0`	none	3	30	162
		composite embedding, KL 1	`emb_040`, `graph_reg_lambda=1`	none	3	30	162
	width	hidden 360, table, no penalty	`hidden_channels=360`, `graph_reg_lambda=0`	none	3	30	162
		hidden 360, table, KL 1	`hidden_channels=360`, `graph_reg_lambda=1`	none	3	30	162
		hidden 360, composite, no penalty	`emb_w360_041`, `graph_reg_lambda=0`	preprocessor rematched (h = 644)	3	30	162
		hidden 360, composite, KL 1	`emb_w360_041`, `graph_reg_lambda=1`	preprocessor rematched (h = 644)	3	30	162
total				80		4,806	

Four code changes, shipped with the plan. The k-hop support is one function shared by both mechanisms, `khop_reach`: the boolean reachability within k steps, a dense $6{,}608^2$ matrix per graph computed once at construction by repeated boolean products; the mask uses it as its support with every self-loop kept, and the prior row-normalizes it as its target with the one-hop diagonal in place of the reach's own (an even walk on an undirected graph returns home, which says nothing about the graph). The first version cleared the diagonal instead; the real-size smoke caught it, since that emptied 6,147 of the 6,607 regulatory rows that the one-hop target constrains to self, and it was fixed before any k-hop job ran. The symmetric prior is one flag in the adjacency normalizer: write $A[j, i]$ beside $A[i, j]$ before row normalization, so the directed graphs get the treatment the mask gives them today. The directed mask is the removal of one line, the transpose write in the mask builder, behind a flag that defaults to today's behavior so no existing config changes meaning. The width arms need the composite preprocessor re-matched to the table's parameter count at 360. Multi-layer placement needs no code: the mask takes a list of layers and the prior's `graph_reg_layer` already accepts a list. The flags default to the behavior every earlier run had, and `tests/torchcell/models/test_cgt_graph_entry.py` pins both the defaults and the new targets on a hand-built directed four-gene graph: a directed one-hop prior, a symmetrized one-hop mask with self-loops, the two- and three-hop reach, the stored self-loops surviving into the k-hop and symmetric targets, and the identity of the mask's support and the prior's support off the diagonal. Every arm's config and overrides in Table 8 compose under Hydra to the intended keys (checked for all 26 distinct combinations), and the plan script writes the Delta submitter, `experiments/025-solid-growth/scripts/delta_submit_round2.sh`, one `sbatch` per run in six chains with round 1b first in every chain. A chain is a stagger, not a queue: Slurm's `after:<id>+30` releases a job 30 minutes after the previous one starts, so that six 554 GB stage-ins never begin on one Lustre path at once, and a later round waits on the scheduler like any other job once its predecessor is running.

The figures the rounds fill. Figure 4 is the wireframe. The first row revises Figure 2: the ladder runs to $\lambda = 100$, the random-graph column moves to $\lambda = 1$, and the checkpoints fill the crosses in the divergence panel. The second row is new and answers reach and direction: one column per mechanism and hop count, from no penalty through the prior at one, two and three hops and its symmetric variant to the mask at one, two and three hops and its directed variant, with the three readings and the stars as in Figure 2a, and the training curves move here from the first figure. The third row holds placement, the budget, and the representation-by-width panel. The two checkpoint panels the first design promised, discovery of edges outside a head's prior and overruling of cataloged edges by the digenic interaction, need no new runs and are scored on the best-Pearson checkpoints already on Delta once they are pulled.

What is not in the rounds, and why. The mask on all eight layers is left out: the placement question is where a graph block belongs, not whether the model can survive having no free layer. The 010 checkpoints and the cosine-schedule runs stay out; their schedule and normalizer differ, and the constant-rate protocol now has its own anchors. Seeds beyond three are not added: at three seeds the epoch-29 reading already separates the arms that differ by 0.01 or more, and the readings that do not separate (max over epochs, the loss-selected epoch) are limited by epoch-to-epoch noise of 0.005 to 0.010 within a run, which more seeds do not remove.

What the rounds measured. Sixty-five of the eighty runs finished; the fifteen that did not are the width-360 arms and the prior on layers 1 to 4, below. Every number is the epoch-29 reading unless stated, mean ± sd over three seeds, with the paired difference against no penalty on the same seeds (Tables 3, 5, 6).

Reach. Longer reach does not help either mechanism. The prior at $\lambda = 1$ reads 0.438 ± 0.006 toward the two-hop target and 0.429 ± 0.012 toward the three-hop target against 0.447 ± 0.001 toward one hop; the paired gain over no penalty shrinks from +0.022 to +0.013 ($p = 0.05$) and +0.004 (mixed by seed), and at the best epoch both k-hop arms sit at 0.442, below no penalty's 0.448 on five seeds of six. The directed mask reads 0.416 ± 0.012, 0.419 ± 0.006 and 0.417 ± 0.003 at one, two and three hops, all at or below no penalty, as the symmetric one-hop mask was. On the seven undirected graphs two hops already reach 60 to 70 percent of all pairs (Table 7), so these arms mostly measure a target close to uniform, and the prior toward it loses about what the uniform part of the target adds.

Direction. Neither mechanism moves toward the other. The prior toward symmetrized targets reads 0.439 ± 0.010 (+0.022, +0.019, +0.002 over no penalty by seed), 0.008 below the directed prior in the mean with one seed carrying the drop; the directed mask reads 0.416 ± 0.012 (−0.002, −0.001, −0.023), not above the symmetric mask's 0.421. The soft-versus-hard gap is therefore not a directed-versus-undirected gap on the regulatory and TFLink heads.

Placement. The prior at $\lambda = 1$ on layers 3 to 4 reads 0.445 ± 0.007 (+0.021, $p = 0.039$), the same as on layer 1, and on layers 1 to 2 0.437 ± 0.009 (+0.012, $p = 0.15$). The mask on layers 1 to 2, 3 to 4 and 1 to 4 reads 0.419, 0.415 and 0.426, at no penalty like the single-layer mask. Where the graphs enter does not change which mechanism helps; the three-seed spread (0.007 to 0.009) does not separate layer 1 from layers 3 to 4.

Budget. At 60 epochs the unpenalized model keeps declining and the prior holds (`experiments/025-solid-growth/results/graph_reg_sweep_history.csv`; the tables read these runs at epoch 29 and their max over 60 epochs). At epoch 59 no penalty is at 0.391, 0.404 and 0.405 by seed, the mask at 0.395, 0.411 and 0.387, the prior at $\lambda = 1$ at 0.433, 0.437 and 0.433, so the fixed-budget gap grows from 0.02 at epoch 29 to 0.03 at epoch 59; the no-penalty and mask maxima stay at epochs 14 to 18, the prior's at 20, 23 and 46. These runs also give a replicate reading: at epoch 29 the 60-epoch runs, the same config and seed as the 30-epoch runs with only `max_epochs` changed and no schedule,

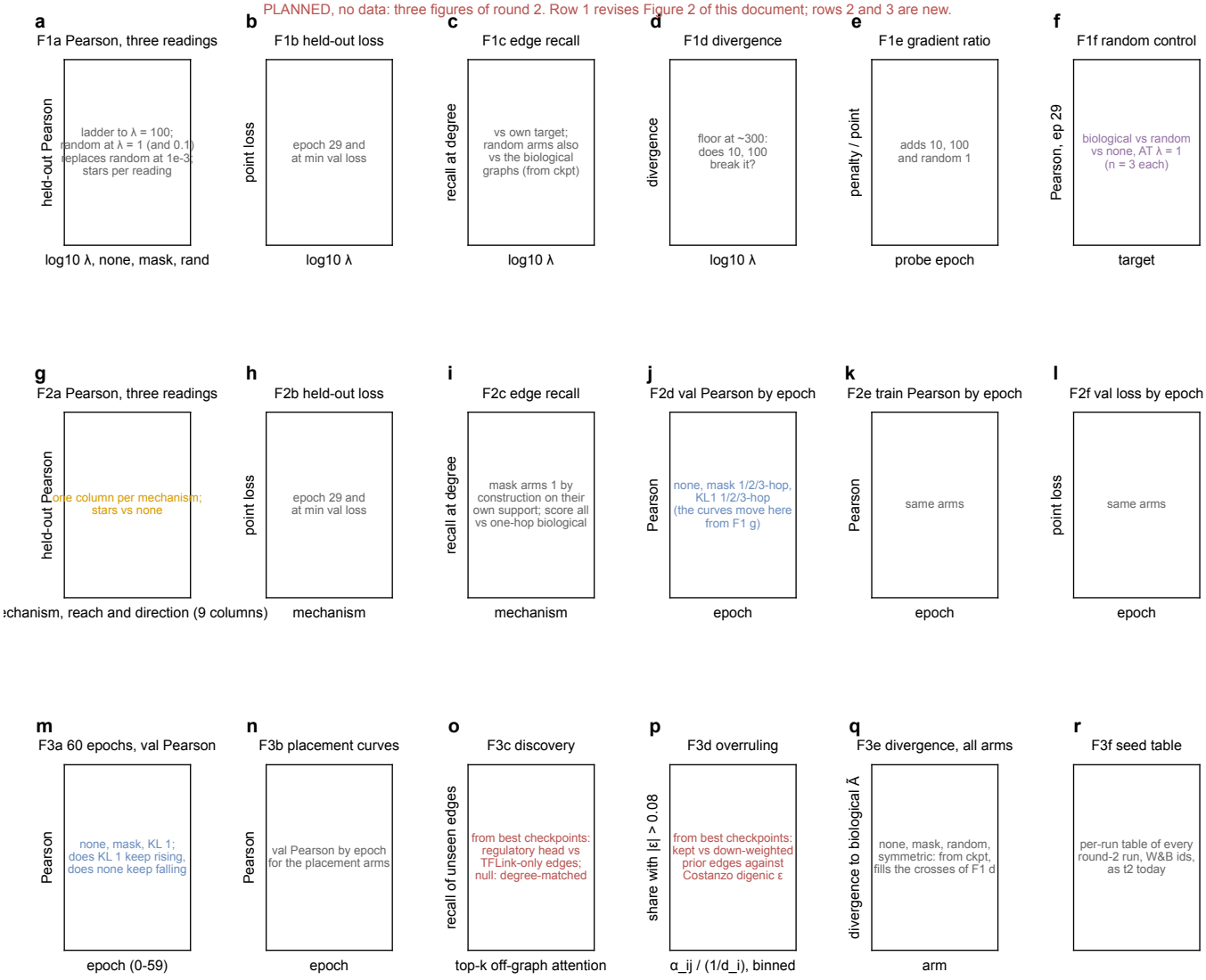


Figure 4: Wireframe of the three planned figures, no data. Row 1 (F1) is Figure 2 extended to $\lambda = 100$ with the random control at $\lambda = 1$, filled by round 1b; row 2 (F2) is the mechanism comparison, one column per way the graphs enter, with the training curves, filled by round 2; row 3 (F3) is placement, the 60-epoch budget, and representation by width, filled by round 3. Generated by `experiments/025-solid-growth/scripts/graph_reg_round2_plan.py`.

read 0.413 to 0.419 with no penalty against 0.425 on all three 30-epoch seeds, 0.410 to 0.428 under the mask against 0.418 to 0.423, and 0.438 to 0.447 at $\lambda = 1$ against 0.446 to 0.448. A same-seed replicate therefore moves by up to 0.011 at epoch 29, the size of every difference among the arms below $\lambda = 10^{-2}$ and of the mask's distance from no penalty, and smaller than the 0.02 the strong prior adds. Hypothesis (untested): the replicate spread is the non-determinism of the four-GPU run (unordered reductions and non-deterministic kernels), since nothing else in the two configs differs.

Representation. With the composite sequence embedding (`emb_040`) and no penalty the model did not train: validation Pearson stays within 0.02 of zero at every epoch on all three seeds (training Pearson 0.003 at epoch 29), and 5 to 11 epochs per seed log it as NaN, an undefined Pearson from constant predictions (Table 6 reads these at the finite epochs). With the prior at $\lambda = 1$ the same model trains, slowly: 0.158 to 0.178 at epoch 0 and 0.421 ± 0.007 at epoch 29, still rising at the last epoch (max at epochs 28 and 29), against 0.447 ± 0.001 for the gene table. The earlier reading that the composite embedding gave +0.016 over the table was made with the joint fitness head (`embfit_035`); without it the composite model needs the prior to leave the origin at all. Hypothesis (untested): the penalty's gradient, 10^3 times the point loss's at these weights, gives the attention a structure the point loss cannot find from the composite input on its own; the per-parameter-group probe of Section 5 would show where it lands.

Width, and the prior on four layers. All twelve width-360 runs (Delta jobs 22462030 to 22462041) and the three runs of the prior on layers 1 to 4 (22462012 to 22462014) failed at the first training step with CUDA out of memory on the 44 GB A40, every rank, so they have no reading. The plan's memory estimate for width was wrong: the attention of the regularized layer is not the whole budget once the hidden width doubles, and four regularized layers exceed the card at width 180. Running these arms means a smaller per-GPU batch with gradient accumulation, which changes the protocol, or a larger card; Delta's A100 nodes are 40 GB and do not help.